MATH 211 Analysis III — Problem Session 4

Problem 1

(a) Let $f(x,y) = \dfrac{xy}{x^2+y^2}$ for $(x,y) \neq (0,0)$. Show that both iterated limits at $(0,0)$ exist and are equal, but $\lim\limits_{(x,y)\to(0,0)} f(x,y)$ does not exist.

(b) Let $g(x,y) = x \sin \dfrac{1}{y}$ for $y \neq 0$. Show that $\lim\limits_{(x,y)\to(0,0)} g(x,y)$ exists, but one of the iterated limits does not exist.

Solution. (a) For $x \neq 0$,

$$\lim_{y\to 0} \frac{xy}{x^2+y^2} = \frac{0}{x^2} = 0 \;\Rightarrow\; \lim_{x\to 0}\lim_{y\to 0} \frac{xy}{x^2+y^2} = 0.$$

For $y \neq 0$,

$$\lim_{x\to 0} \frac{xy}{x^2+y^2} = \frac{0}{y^2} = 0 \;\Rightarrow\; \lim_{y\to 0}\lim_{x\to 0} \frac{xy}{x^2+y^2} = 0.$$

Along $y = 0$:

$$\lim_{(x,0)\to(0,0)} \frac{xy}{x^2+y^2} = \lim_{x\to 0} \frac{0}{x^2} = 0.$$

Along $y = x$:

$$\lim_{(x,x)\to(0,0)} \frac{xy}{x^2+y^2} = \lim_{x\to 0} \frac{x^2}{2x^2} = \frac{1}{2}.$$

The limits along two paths differ, so the double limit does not exist.

(b) Since $\left|\sin \frac{1}{y}\right| \leq 1$,

$$0 \leq \left| x \sin \frac{1}{y} \right| \leq |x|, \qquad \lim_{(x,y)\to(0,0)} |x| = 0,$$

so by the squeeze theorem

$$\lim_{(x,y)\to(0,0)} x \sin \frac{1}{y} = 0.$$

For fixed $x \neq 0$, $\lim\limits_{y\to 0} x \sin \dfrac{1}{y}$ does not exist (it oscillates between $-x$ and x), so

$$\lim_{x\to 0}\lim_{y\to 0} x \sin \frac{1}{y} \quad \text{does not exist.}$$

Problem 2

Show, using the ϵ–δ definition, that $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^2$ is continuous at every $a \in \mathbb{R}$.

Solution. Let $\epsilon > 0$ be given and $a \in \mathbb{R}$. We find a $\delta > 0$ such that $|f(x) - f(a)| < \epsilon$ whenever $|x - a| < \delta$.

$$|f(x) - f(a)| = |x^2 - a^2| = |x + a||x - a|$$

is the expression we wish to bound above by ϵ. Observe that if $|x-a| < \delta$, we have no information about $|x + a|$. The δ may depend on a but it can't depend on x. To solve this issue: Suppose

$\delta \leq 1$. We observe that $|x| = |x-a+a| \leq |x-a|+|a| \leq 1+|a|$, which is a bound on $|x|$ depending only on a. It follows that $|x+a| \leq |x|+|a| \leq 2|a|+1$. Then,

$$|f(x) - f(a)| = |x^2 - a^2| = |x+a||x-a| < (2|a|+1)\delta.$$

Choosing $\delta = \min\{1, \frac{\epsilon}{2|a|+1}\}$ gives $|f(x) - f(a)| < \epsilon$ whenever $|x-a| < \delta$.

Question. Why is taking the minimum necessary?

Problem 3

(a) Show that $f : \mathbb{R}^2 \to \mathbb{R}$, $f(x,y) = 2x + 3y$ is uniformly continuous.

(b) Show that $h : \mathbb{R} \to \mathbb{R}$, $h(x) = x^2$ is not uniformly continuous.

Solution. (a) Let $\epsilon > 0$ be given. We find a $\delta > 0$ (only depending on ϵ) satisfying $|f(x)-f(y)| < \epsilon$ for any $x = (x_1, x_2)$, $y = (y_1, y_2)$ whenever $\|x - y\| < \delta$. Observe that $|x_1 - y_1|$ and $|x_2 - y_2|$ are both bounded above by $\|x - y\|$. Thus,

$$|f(x)-f(y)| = |(2x_1+3x_2)-(2y_1+3y_2)| = |2(x_1-y_1)+3(x_2-y_2)| \leq 2|x_1-y_1|+3|x_2-y_2| \leq 5\|x-y\|.$$

Now choosing $\delta = \frac{\epsilon}{5}$ gives $|f(x) - f(y)| < \epsilon$.

Solution. Suppose h is a uniformly continuous function. Then for given $\epsilon = 1$ there exists $\delta > 0$ such that $|h(x) - h(y)| < \epsilon$ whenever $|x - y| < \delta$. Choose $n \in \mathbb{N}$ such that $\frac{1}{n} < \delta$ (Why can we always choose such n?). Let $x = n$ and $y = n + \frac{1}{n}$, so $|x - y| = \frac{1}{n} < \delta$. It follows that $|h(x) - h(y)| = |n^2 - (n^2 + 2 + \frac{1}{n^2})| = |2 + \frac{1}{n^2}|$ which is obviously greater than 1, which is a contradiction. Thus, h is not uniformly continuous.

Exercise. Show that $h(x) = x^2$ is uniformly continuous on $[0, 1]$.

Problem 4

Recall: if f is continuous, then f^{-1}(open) is open and f^{-1}(closed) is closed. Use this to show:

(a) $A = \{(x,y) \in \mathbb{R}^2 : x^2 + y^2 < 4,\ xy > 1\}$ is open.

(b) $B = \{(x,y) \in \mathbb{R}^2 : y \geq x^2,\ x + y \leq 2\}$ is closed.

Solution. (a) Let $f(x,y) = x^2 + y^2$ and $g(x,y) = xy$. Then $A = f^{-1}((-\infty, 4)) \cap g^{-1}((1, +\infty))$ (Why?). Since $(-\infty, 4)$ and $(1, \infty)$ are open sets, $f^{-1}((-\infty, 4))$ and $g^{-1}((1, \infty))$ are open, so their intersection A is open.

(b) Let $h(x,y) = y - x^2$ and $w(x,y) = x + y$. Then $B = h^{-1}([0, +\infty)) \cap w^{-1}((-\infty, 2])$. Since $[0, +\infty)$ and $(-\infty, 2]$ are closed sets, both pre-images are closed, so their intersection B is closed.

Showing the functions f, g, h, w are continuous is left as an exercise.

Problem 5

Let f be a continuous function.

(a) Is the preimage of a compact set under f compact? Prove it or give a counterexample.

(b) Is the preimage of a connected set under f connected? Prove it or give a counterexample.

Solution. (a) No. Consider $f(x) = 1$ and the compact set $\{1\}$. Its pre-image is $\mathbb{R}$, which is not compact.

(b) No. Consider $g(x) = x^2$ and the pre-image of $[4, 9]$, which is $[-3, -2] \cup [2, 3]$ and not connected.

Problem 6

Let $C = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 1\}$ and $P = (3, 0)$. Show that there is a point of C closest to P (use the Extreme Value Theorem), and find it.

Solution. The distance $d(x, P)$ between any point $x = (x_1, x_2)$ in C and P is $d_P(x) = \sqrt{(x_1 - 3)^2 + x_2^2}$. Noting that $x_2^2 = 1 - x_1^2$ we get $d_P(x) = \sqrt{(x_1 - 3)^2 + 1 - x_1^2}$, which is a continuous function of x because square root function is continuous and composition of continuous functions is again continuous. Now, by the Extreme Value Theorem the maximum (or minimum) value of $d_P(x)$ exists because C is compact. In order to find the maximum (or minimum) value of $d_P(x)$, use your knowledge from analysis 1 (find critical points etc.)

Problem 7

Using polar coordinates $x = r\cos\theta$, $y = r\sin\theta$, compute

(a) $\lim\limits_{(x,y)\to(0,0)} \dfrac{x^3 + y^3}{x^2 + y^2}$,

(b) $\lim\limits_{(x,y)\to(0,0)} (x^2 + y^2)\ln(x^2 + y^2)$.

Solution. (a) $x = r\cos\theta$, $y = r\sin\theta$.

$$\frac{x^3 + y^3}{x^2 + y^2} = \frac{r^3(\cos^3\theta + \sin^3\theta)}{r^2} = r(\cos^3\theta + \sin^3\theta).$$

Since $|\cos^3\theta + \sin^3\theta| \le 2$ (by triangle inequality),

$$0 \le \left|\frac{x^3 + y^3}{x^2 + y^2}\right| \le 2r$$

so by the squeeze theorem

$$\lim_{(x,y)\to(0,0)} \frac{x^3 + y^3}{x^2 + y^2} = 0.$$

(b) Since $x^2 + y^2 = r^2$, let $t = r^2$. Then $t \to 0^+$ and

$$\lim_{(x,y)\to(0,0)} (x^2 + y^2)\ln(x^2 + y^2) = \lim_{t\to 0^+} t\ln t = \lim_{t\to 0^+} \frac{\ln t}{1/t} = \lim_{t\to 0^+} \frac{1/t}{-1/t^2} = \lim_{t\to 0^+} (-t) = 0.$$